

DSA8045 – Applied Analytics Assignment 1 (30% of module mark)

Task Overview

Working in groups of 5–6 students, you will design, develop and present an interactive dashboard using R Shiny. The purpose of this assessment is to demonstrate your understanding of data visualisation principles, interactive dashboard design, and communication of insights using real-world data.

Each group will be given one dataset from the list of approved datasets provided on Canvas. Using this dataset, your group will create a dashboard that allows users to explore, filter and interpret the data through interactive visualisations and summary outputs. The dashboard should be designed for a non-technical audience and should clearly communicate key findings and trends within the data.

The project is intended to develop both technical and collaborative skills. As part of the assessment, students are expected to contribute to the planning, coding, design, interpretation and presentation of the dashboard. The dashboard should demonstrate effective use of R Shiny functionality, appropriate visualisation techniques using ggplot2, and clear communication of analytical findings.

Your dashboard should demonstrate good dashboard design principles, including readability, accessibility, logical structure and ease of navigation. Interactive elements should enhance the user experience and support exploration of the data rather than simply displaying static plots.

Your dashboard must include:

- A clear title and introduction section
- A minimum of three dashboard tabs/pages
- At least three interactive visualisations
- At least two user input controls (e.g. dropdown menus, sliders, checkboxes)
- At least two reactive features that dynamically update outputs based on user input
- A conclusions or recommendations section
- Appropriate use of ggplot2 visualisations
- Clear labels, legends and axis titles throughout

The dashboard should include a variety of visualisation types where appropriate (e.g. bar charts, scatterplots, boxplots, line charts, maps or tables). Students are encouraged to focus on clarity, usability and interpretation of findings rather than excessive technical complexity.

Each group will deliver a 20 minute in-class presentation demonstrating the functionality of the dashboard and discussing the insights obtained from the data.

The presentation should include:

- an overview of the dataset
- explanation of dashboard structure and design choices
- demonstration of interactive features
- discussion of key findings and conclusions
- reflection on how the dashboard supports user understanding

All group members are expected to contribute to the presentation.

Submission Deadline and Details

This piece of coursework will consist of 30% of each student's final mark. Assessment is based on an R Shiny dashboard, peer assessment, and delivery of the presentation of the dashboard.

Students must submit:

- R Shiny code
- HTML output
- Peer assessment form - submitted individually

R Shiny dashboard R code/HTML submission and peer assessment form deadline:
Sunday 5th July 23:59

Presentation of the dashboard in groups:
Monday 6th July 10:00

Late Submission of Assessed Work

Coursework submitted after the deadline will be penalised at the rate of 5 percentage points deducted from the total mark awarded for that component for each calendar day late up to a maximum of five calendar days, after which a mark of zero shall be awarded. This excludes University closure days. Exemptions shall be granted to the above only if there are exceptional circumstances. Extensions to deadlines shall be proportionate to the impact of the exceptional circumstances.

Assessment

The group project will be assessed based on the in-person presentation (40%), the accompanying R code/HTML (40%), and peer assessment of individual contribution (20%).

Peer assessment weightings or scores may be moderated in exceptional circumstances to ensure fairness and consistency.

Each team member is expected to contribute throughout the planning, code creation and interpretation of the R Shiny dashboard and presentation. A student's contribution to the project will be taken into account via peer assessments. A small element of the assessment acknowledges individual contributions. In order to do this, all students will be asked to allocate a theoretical set of marks amongst individuals in the group, according to the contribution level they perceive for each team member. You may wish to develop group criteria for the allocation of these marks, but each student will be expected to submit their preferred allocation of marks on an individual basis by the deadline above.

The criteria used to assess the this piece of coursework includes:

Assessment Component	Weighting	
Dashboard Development & Design	40%	Effective use of R Shiny, reactive components, user controls, code quality, functionality, visualisation design and communication of insights.
Presentation & Communication	40%	Clear explanation of the dashboard, communication of findings, professional delivery and engagement during the presentation.
Peer Assessment & Contribution	20%	Individual contribution to planning, coding, analysis, dashboard development and presentation.

Plagiarism

The university policy on plagiarism is available at:

<https://www.qub.ac.uk/directorates/sgc/learning/LearningResources/>

Remember plagiarism is a serious offense - ensure that you use your own words even when referencing a source. Plagiarism means presenting work or ideas from another source as your own, without appropriate acknowledgement and references. For this assignment this includes unacknowledged and undeclared use of generative AI tools, in either the code or the report. Also reference libraries, tutorials, or code that you used.